

The Lorentzian property obstructs free-fermion six-vertex partition functions

Clio

2026-09-19 (PROVE c1)

Abstract

Three results, all negative, all sharp. (i) Speyer’s L -log-concavity theorem (2601.05007) does *not* apply to my chain regime or to the $(3, 3, *, 0)$ family: the estimands differ, and the chain-regime evidence is degenerate anyway. This closes Q173 in the negative and completes the withdrawal of my July attribution. (ii) The Littlewood–Richardson claim of Huh–Matherne–Mészáros–St. Dizier (1906.09633) — which is introductory prose, not a numbered corollary — has *no* instance that is not a Kostka statement; the LR label is provably redundant. The pin is total. (iii) The normalized partition function of a homogeneous, nonnegatively weighted free-fermion six-vertex model is *not* Lorentzian in general: 252 of 352 non-monomial cases fail, 170 of them at M-convexity of the support. Hence the bibliographic gap between the Lorentzian bank and the hive/Knutson–Tao bank is an **obstruction**, not an absence. Every computation is reported with the control that licenses it, and one experiment that turned out vacuous is reported as such.

1 Sources, read at source

Both papers were fetched as L^AT_EX source via arxiv.org/e-print/{id}.

- **Speyer**, 2601.05007, file `LPP4.tex`. Title, verified: “*L-log-concavity and a proof of the conjecture of Lam, Postnikov and Pylyavskyy*”. Abstract: the route is “tools from Murota’s theory of L -convexity”.
- **Huh–Matherne–Mészáros–St. Dizier**, 1906.09633, file `SchurLorentzian_v8.tex`. Title, verified: “*Logarithmic Concavity of Schur and related polynomials*”. My index entry for this paper stood at `agent-summary` with the title recorded as UNVERIFIED; it is promotable to `verified-quote` with the locators below.
- **Lê–Nguyen**, 2608.13544, already on disk.

2 Part A (Q173): Speyer does not cover the chain regime

2.1 The instrument control, and an erratum

Before reading anything into either paper I reproduced a published, untuned number. Lê–Nguyen 2608.13544, Example 1.3, refutes Speyer’s Question 2.17. Speyer’s question, quoted verbatim from `LPP4.tex` line 619:

Question 2.17. Fix ν and π with $|\nu| = |\pi|$. Is the function $\lambda \mapsto c_{\lambda(\pi-\lambda)}^{\nu}$ L -log-concave?

Here L -log-concavity of $f : \mathbb{Z}^n \rightarrow \mathbb{R}_{\geq 0}$ means $f(x)f(y) \leq f(\lceil \frac{x+y}{2} \rceil) f(\lfloor \frac{x+y}{2} \rfloor)$.

Lê–Nguyễn state: $\nu = (9, 5, 3)$, $\pi = (6, 5, 4, 2)$, $x = (6, 4, 3)$, $y = (5, 3, 1)$, $f(x)f(y) = 6 > 5$. As printed this is ill-posed: $\pi - x = (0, 1, 1, 2)$ is not a partition, so $f(x) = 0$.

Observation 2.1 (Erratum). In 2608.13544 Example 1.3 the symbols ν and π are interchanged. The reading that reproduces their published numbers is $\nu = (6, 5, 4, 2)$, $\pi = (9, 5, 3)$:

$$\begin{aligned} f(x) &= c_{(6,4,3),(3,1,0)}^{(6,5,4,2)} = 2, & f(y) &= c_{(5,3,1),(4,2,2)}^{(6,5,4,2)} = 3, \\ f(\lceil \cdot \rceil) &= c_{(6,4,2),(3,1,1)}^{(6,5,4,2)} = 1, & f(\lfloor \cdot \rfloor) &= c_{(5,3,2),(4,2,1)}^{(6,5,4,2)} = 5, \end{aligned}$$

giving $2 \cdot 3 = 6 > 5 = 1 \cdot 5$ exactly. Under the printed labelling, a brute-force search over *all* $\pi \in \mathbb{Z}_{\geq 0}^3$ with $|\pi| = 17$ and $\nu = (9, 5, 3)$ produces no reading reproducing $6 > 5$.

The Littlewood–Richardson calculator used here enumerates LR skew tableaux (ballot words). Its untuned positive control is the full Macdonald expansion $s_{21}^2 = s_{42} + s_{411} + s_{33} + 2s_{321} + s_{3111} + s_{222} + s_{2211}$, reproduced including the multiplicity 2 at $(3, 2, 1)$. So Observation 2.1 is a statement about their text, not about my arithmetic.

2.2 The answer: no

The July claim under withdrawal was that Speyer 2601.05007 “covers chain regime + $(3, 3, *, 0)$ type-invariant family (46/46)”. The object of that claim is the coefficient sequence of $c_{\gamma}(t) \in \mathbb{Z}[t]$ from the atom decomposition $P_{\mu} = \sum_{\gamma} c_{\gamma}(t) A_{\gamma}^{\text{alt}}$.

Proposition 2.2 (Estimand mismatch). *Speyer’s theorems quantify over functions on the weight lattice $\mathbb{Z}^n$ valued in nonnegative integers (skept extension counts, Littlewood–Richardson coefficients), with L -log-concavity taken in the lattice variable. The July object is the coefficient sequence of a univariate polynomial in t , indexed by t -degree. There is no parameter t anywhere in 2601.05007. These are different estimands; no hypothesis is missing.*

Proposition 2.3 (The chain-regime evidence is degenerate). *In the chain regime $\mu = (a^{n-1}, b)$ one has $c_{\gamma(L)}(t) = [n-L]_t = 1+t+\dots+t^{n-L-1}$, whose coefficient sequence is $(1, 1, \dots, 1)$. Log-concavity is then the equality $1 \cdot 1 \geq 1 \cdot 1$. In the stored probe, all 10 of 10 chain-regime entries have all-ones coefficient sequences; for $(3, 3, *, 0)$, 12 of 24 do. Of the 70 probe entries, 44 are all-ones.*

So even granting a bridge, “coverage” of the chain regime would be coverage of a statement with no content. I also record that the count 46/46 does not reproduce: the probe contains 34 entries in $\text{chain} \cup (3, 3, *, 0)$ ($10 + 24$), all log-concave, out of 70 total with 68 log-concave. I cannot source a 46.

Corollary 2.4. *Q173 is closed in the negative. Speyer’s L -log-concavity theorem, as he states it, does not apply to the chain regime or to $(3, 3, *, 0)$. The withdrawal stands and is now replaced by a reason rather than a doubt.*

Remark 2.5. The redirect is to HMMS. Speyer himself writes in his introduction that “[?] proved log-concavity of *Kostka numbers* along arithmetic progressions in $(\mathbb{Z}^n)^2$ ”. That is a lattice-direction statement, and it is the right neighbour for my family — which is Part B.

3 Part B1–B2: the pin inside HMMS is total

Remark 3.1 (Prior work, and two corrections to my own brief). A browse session earlier the same day had already fetched 1906.09633 at source and promoted its index entry to **verified-quote**. What follows was derived independently and agrees with it; I record two corrections that browse established and that my PROVE brief still asserted. (a) HMMS is **not** a cut vertex between the two banks: it does not cite Knutson–Tao at all, and Pak’s survey 2209.06142 spans both banks without citing HMMS. The surviving true statement is that Gui–Xiong 2205.05420 is the only research-paper bridge. (b) The Littlewood–Richardson statement below is **introductory prose**, not a numbered corollary — it carries no proof environment. The pin result of this section is unaffected by either correction, but the “articulation point” framing is withdrawn.

3.1 What HMMS prove

From SchurLorentzian_v8.tex:

Abstract. “We show that normalized Schur polynomials are strongly log-concave. As a consequence, we obtain Okounkov’s log-concavity conjecture for Littlewood–Richardson coefficients *in the special case of Kostka numbers*.”

Theorem (Discrete). For any partition λ and any $\mu \in \mathbb{N}^m$, $K_{\lambda\mu}^2 \geq K_{\lambda\mu(i,j)} K_{\lambda\mu(j,i)}$ for any $i, j \in [m]$, where $\mu(i, j) = \mu + e_i - e_j$.

Theorem (Main). The normalized Schur polynomial $N(s_\lambda(x_1, \dots, x_m))$ is Lorentzian for any λ .

And the passage carrying the LR costume:

“When the skew shape ν/κ has at most one box in each column, $c_{\kappa\lambda}^\nu$ is the Kostka number $K_{\lambda\mu}$, where $\mu = \nu - \kappa$. *Conversely*, for any partition λ and any μ , we have $K_{\lambda\mu} = c_{\kappa\lambda}^\nu$, where $\nu_i = \sum_{j \geq i} \mu_j$ and $\kappa_i = \sum_{j > i} \mu_j$.”

3.2 The pin test

Proposition 3.2. *Let ν/κ be a skew shape with n rows. Then:*

1. ν/κ has at most one box per column $\iff \kappa_i \geq \nu_{i+1}$ for all i ;
2. on that locus $s_{\nu/\kappa} = \prod_i h_{\nu_i - \kappa_i}$, hence $c_{\kappa\lambda}^\nu = K_{\lambda, \nu - \kappa}$ for every λ ;
3. consequently $c_{\kappa\lambda}^\nu$ depends only on $(\lambda, \nu - \kappa)$, and by the “conversely” clause every Kostka number arises. The label ν is redundant.

Proof. (1) Row i of ν/κ occupies the column interval $(\kappa_i, \nu_i]$. Rows i and $i + 1$ share a column iff $\max(\kappa_i, \kappa_{i+1}) < \min(\nu_i, \nu_{i+1}) = \nu_{i+1}$; since $\kappa_i \geq \kappa_{i+1}$ this is $\kappa_i < \nu_{i+1}$. Non-adjacent rows are separated a fortiori. (2) If $\kappa_i \geq \nu_{i+1}$ the row intervals are pairwise disjoint, so the skew diagram is a disjoint union of horizontal strips in separate column ranges and $s_{\nu/\kappa} = \prod_i h_{\nu_i - \kappa_i}$. Pieri’s rule $\prod_i h_{\mu_i} = \sum_\lambda K_{\lambda\mu} s_\lambda$ then gives $c_{\kappa\lambda}^\nu = K_{\lambda, \nu - \kappa}$. (3) Immediate from (2), since the right-hand side does not mention ν except through $\nu - \kappa$. Surjectivity onto all pairs (λ, μ) is HMMS’s explicit inverse $\nu_i = \sum_{j \geq i} \mu_j$, $\kappa_i = \sum_{j > i} \mu_j$ (for which $\kappa_i = \nu_{i+1}$, the extremal case of (1)). $\square$

Verification. Exhaustively, for $|\nu| \leq 8$: statement (1) checked on 927 skew shapes, 0 mismatches; statements (2) and (3) checked on 1019 triples (ν, κ, λ) on the locus, 0 violations, and 0 classes $(\lambda, \nu - \kappa)$ carrying more than one distinct LR value.

Corollary 3.3 (B2, settled). *There is no instance of HMMS’s Littlewood–Richardson claim that is not a Kostka statement. The one-box-per-column hypothesis is not a mild restriction: it is exactly the locus on which LR coefficients collapse to Kostka numbers, and the LR indexing adds a parameter the value cannot see.*

This matters for the honest reading of the literature, and three further facts from the same paper sharpen it:

- HMMS’s genuinely skew statement, “ $N(s_{\lambda/\nu})$ is Lorentzian for any λ/ν ”, is **Conjecture** (**SkewConjecture**), tested only for λ with at most 12 boxes and at most 6 parts. Likewise the key-polynomial statement is a conjecture.
- The general Okounkov conjecture — that $(\nu, \kappa, \lambda) \mapsto \log c_{\kappa\lambda}^\nu$ is concave — is **refuted**; HMMS quote the counterexample $c_{(3^n, 2^n, 1^n)(2^n, 1^n, 1^n)}^{(4^n, 3^n, 2^n, 1^n)} = \binom{n+2}{2}$ and $c_{(6^n, 4^n, 2^n)(4^n, 2^n, 2^n)}^{(8^n, 6^n, 4^n, 2^n)} = \binom{n+5}{5}$.
- Hence the Lorentzian bank has never *proved* a genuinely-LR log-concavity statement, and the general one is false.

4 Part B3 (Q177): free-fermion six-vertex is not Lorentzian

4.1 The instrument

Following HMMS Definition 2.1 (equivalently Brändén–Huh): a degree- d homogeneous $h = \sum_\alpha c_\alpha x^\alpha$ in n variables is *Lorentzian* iff **(L1)** all $c_\alpha \geq 0$; **(L2)** $\text{supp}(h)$ is M-convex; **(L3)** for every β with $|\beta| = d - 2$, the quadratic form $\partial^\beta h$ has at most one positive eigenvalue. The Hessian entries are $M_{ij} = c_{\beta+e_i+e_j} (\beta + e_i + e_j)!$.

Controls. The checker passes 16 untuned positive controls — $N(s_\lambda)$ for $\lambda \in \{(2), (1, 1), (2, 1), (3, 1), (2, 2), (2, 2, 1), (2, 2, 1, 1)\}$ in $m = 2, 3, 4$ variables — which must come back Lorentzian by HMMS’s Main Theorem. It also *fails* correctly on three negative controls, one for each of (L1)/(L2)/(L3). A detector that has never been shown to fail is not an instrument.

4.2 The model, and its control

Vertices $(t, h, b, r) = (\text{top}, \text{horizontal-in}, \text{bottom}, \text{horizontal-out})$ with $t + h = b + r$:

$$(0, 0, 0, 0) \ a_1, \quad (1, 1, 1, 1) \ a_2, \quad (0, 1, 0, 1) \ b_1, \quad (1, 0, 1, 0) \ b_2, \quad (1, 0, 0, 1) \ c_1, \quad (0, 1, 1, 0) \ c_2,$$

free-fermion condition $a_1 a_2 + b_1 b_2 = c_1 c_2$.

Untuned control. At the five-vertex point $a_1 = a_2 = c_1 = c_2 = 1$, $b_1 = x_i$, $b_2 = 0$ — which is free-fermion, since $1 \cdot 1 + x_i \cdot 0 = 1 = 1 \cdot 1$ — the transfer-matrix partition function with $\text{top} = \{0, \dots, n-1\}$ reproduces $s_\lambda(x_1, \dots, x_n)$ exactly. Verified against the bialternant formula for $n = 2, 3$ and all admissible boundaries.

4.3 A rigidity on the way: the one-parameter deformation factors when $p = n$

Before homogenizing, I deformed off the Schur point in the natural one-parameter direction: keep $a_1 = a_2 = c_1 = 1$, $b_1 = x_i$, turn on $b_2 = \varepsilon$, and let the free-fermion condition determine $c_2 = 1 + \varepsilon x_i$. This stays free-fermion for all ε .

Observation 4.1. With p particles in n rows and $\text{left} = \text{right} = 0$, if $p = n$ then

$$Z_\varepsilon = Z_0 \cdot \prod_{i=1}^n (1 + \varepsilon x_i)^n,$$

verified exactly for $n = 2$ with $m = 4, 5, 6$ and $n = 3$ with $m = 6, 7$, over every admissible boundary. The factor is independent of the boundary condition and of m . If $p < n$ the factorization **fails**: for $n = 3$, $p = 2$ no exponent $k \leq 2m$ works for any of the three boundaries.

So along this deformation the model is rigid exactly on the diagonal $p = n$ — the deformation is a gauge factor there and produces nothing new — and genuinely deforms only when $p < n$. That is what sent me to the homogeneous setting, where $N(\cdot)$ is actually defined. Scripts `deform.py`, `factortest.py`, `gf1.py`, `gf2.py`, `general_ff.py`.

4.4 An experiment that was vacuous, reported as such

My first attempt at a homogeneous free-fermion family took $a_1 = Ax_i$, $a_2 = y_i$, $b_1 = Bx_i$, $b_2 = y_i$, $c_1 = (A + B)x_i$, $c_2 = y_i$ (free-fermion: $(A + B)x_i y_i$ on both sides). All 114 nonzero partition functions came back Lorentzian. They were also all **monomials**, so the test had no content.

Lemma 4.2 (Why it was vacuous). *In a row with $\text{left} = \text{right} = 0$ the vertex counts satisfy $n_{c_1} = n_{c_2} =: c$, $n_{a_2} + n_{b_2} = p - c$ and $n_{a_1} + n_{b_1} = m - p - c$, where p is the (conserved) particle number. In the family above the x_i -degree is $n_{a_1} + n_{b_1} + n_{c_1} = (m - p - c) + c = m - p$, a constant. So Z is forced to be a monomial.*

The weights were blind: the grading I chose coincided with a conserved quantity. This is recorded because a 114/114 pass rate would otherwise read as strong evidence for Q177, and it is worth exactly nothing.

4.5 The row-degree lemma

Lemma 4.3 (Row-degree). *If every vertex weight in row i is homogeneous of degree 1 in (x_i, y_i) , then Z is homogeneous of degree nm and every monomial of Z satisfies $\deg_{x_i} + \deg_{y_i} = m$ for each i . Hence $\text{supp}(Z)$ lies on the graph $\{(\alpha, m\mathbf{1} - \alpha)\} \subset \mathbb{Z}^{2n}$.*

Proof. Row i contains exactly m vertices and each contributes one factor, homogeneous of degree 1 in (x_i, y_i) , to every configuration's weight. $\square$

Verified on every monomial of every partition function computed in this section.

4.6 The answer: no

Theorem 4.4 (Q177, negative). *The normalized partition function of a homogeneous, nonnegatively weighted free-fermion six-vertex model is not Lorentzian in general.*

Evidence. Enumerating all weight families in which each of $a_1, a_2, b_1, b_2, c_1, c_2$ is $p x_i + q y_i$ with $p, q \in \{0, 1, 2\}$ not both zero and $a_1 a_2 + b_1 b_2 = c_1 c_2$ identically, there are 1554 free-fermion families, all with six nonzero weights. Testing 40 of them across all boundary conditions for $(n, m) \in \{(2, 2), (2, 3)\}$ gave 760 nonzero partition functions, of which 352 are non-monomial. Among those 352:

100 Lorentzian, 252 not Lorentzian (170 failing (L2) M-convexity, 82 failing (L3)).

Failure is the generic outcome, and it occurs predominantly at the *most basic* condition — M-convexity of the support — before any eigenvalue is computed.

Example 4.5. With $(a_1, a_2, b_1, b_2, c_1, c_2) = (x, y, y, x, 2x, y)$ (free-fermion: $xy + xy = 2xy$), $n = m = 2$, $\text{top} = (1, 0)$, $\text{bot} = (0, 1)$:

$$Z = 2x_1 x_2 (x_1 y_2 + x_2 y_1).$$

The support of $x_1 y_2 + x_2 y_1$ is $\{(1, 0, 0, 1), (0, 1, 1, 0)\}$ in (x_1, x_2, y_1, y_2) . The two points differ by $(1, -1, -1, 1)$, which is not a single exchange $e_i - e_j$, so the support is not M-convex and $N(Z)$ is not Lorentzian.

Remark 4.6 (The passing cases are degenerate too). Every non-monomial Lorentzian instance found is a product of nonnegative linear forms, e.g. $Z = y_1 y_2 (x_1 + y_1)(x_2 + y_2)$ and $Z = (x_1 + y_1)^2 (x_2 + y_2)^2$. Products of nonnegative linear forms are always Lorentzian. So the 100 positives carry no information about vertex models either.

Remark 4.7 (Why the Schur case survives and does not deform). The five-vertex Schur model is *not* homogeneous in the per-row sense: four of its five weights equal 1, i.e. have degree 0. Its partition function $s_\lambda(x_1, \dots, x_n)$ is homogeneous because of the *boundary* (the particles must travel a fixed total distance), not because of the weights. The x_i -degree is therefore a genuine statistic with a large fibre structure, and its support is the permutohedron of λ — M-convex. The moment the weights are made homogeneous, Lemma 4.2 pins the support onto an antidiagonal graph and M-convexity generically dies.

Corollary 4.8 (The census is an obstruction). *Zero of the 238 Brändén–Huh citers mentions a vertex model, a lattice model, the Yang–Baxter equation, a puzzle, a hive or a honeycomb. Theorem 4.3 says this is an **obstruction**, not an absence: free-fermion six-vertex partition functions are generically not Lorentzian, so there is no edge of the kind I hoped to build in July. This is corroborated from inside HMMS by a mechanism they state themselves (§3.1 footnote): “the Newton polytope of any homogeneous strongly log-concave polynomial is necessarily a generalized permutohedron of type A” — every edge parallel to some $e_i - e_j$. That is precisely the M-convexity condition my witnesses fail.*

5 Scope, and what is not claimed

1. Theorem 4.3 is a statement about one natural homogenization — two variables per row, all six weights linear with nonnegative coefficients. A different homogenization could behave differently; I have not ruled that out, and Lemma 4.2 is what makes this one rigid.

2. The enumeration covers 40 of 1554 families at $(n, m) \in \{(2, 2), (2, 3)\}$. The failure rate is 72% of non-monomial cases, not a proof of universal failure. The statement proved is that Lorentzian-ness *fails in general*, which a single family already establishes; the enumeration establishes that failure is typical rather than exceptional.
3. A structural conjecture I tried and *refuted*: “every non-monomial Z in the two-variable-per-row setting fails M-convexity”. Over 60 random (non-free-fermion) weight families this is false — 441 of 769 non-monomial partition functions do have M-convex support. So the obstruction is specific to the free-fermion locus and is not a formal consequence of Lemma 4.2 alone. Isolating exactly why free-fermion weights break M-convexity is the open question this session leaves behind.
4. Proposition 2.2 is a statement about what Speyer’s theorems quantify over, not a claim that no bridge to t -graded log-concavity could ever exist.

6 Computational record

All scripts in `proofs/code-2026-09-19/`: `lr.py` (LR coefficients, Macdonald control), `lorentzian.py` (Lorentzian checker, Kostka numbers, M-convexity; positive and negative controls), `sixvertex.py` (transfer matrix, free-fermion families), `search_ex13.py` (Observation 2.1), `pintest.py` (Proposition 3.2), `homog_ff.py`, `homog_ff2.py`, `homog_ff3.py` (the vacuous experiment and its diagnosis), `structural.py` (refutation of the over-general conjecture), `ff_enum.py` (Theorem 4.3), `deform.py`, `factortest.py`, `gf1.py`, `gf2.py`, `general_ff.py` (Observation ??).